

# ECN 594: Oligopoly Refresher

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# Plan

1. **Cournot competition (refresher)**
2. Bertrand competition (refresher)
3. Cournot vs Bertrand

## From ECN 532: Oligopoly models

- You covered Cournot and Bertrand in Hector's class
- Today: quick refresher in IO notation
- **New focus:** Market power measurement
  - Connecting oligopoly models to Lerner index
  - When does each model apply?

## Cournot competition: setup

- $n$  firms producing homogeneous goods
- Firms choose **quantities** simultaneously
- Inverse demand:  $P = P(Q)$  where  $Q = \sum_{i=1}^n q_i$
- Constant marginal cost:  $c$
- Firm  $i$  profit:  $\pi_i = P(Q) \cdot q_i - c \cdot q_i$

## Worked example: Cournot duopoly

- **Question:** Inverse demand is  $P = 100 - Q$ . Two symmetric firms with  $MC = 10$ .
- Find equilibrium quantities and price.

## Worked example: Cournot duopoly (solution)

- **Solution:**
- Firm  $i$  profit:  $\pi_i = (100 - q_i - q_j)q_i - 10q_i$
- FOC:  $\frac{d\pi_i}{dq_i} = 100 - 2q_i - q_j - 10 = 0$
- Symmetric:  $q_1 = q_2 = q^*$ , so  $100 - 3q^* = 10$
- $q^* = 30$
- $Q = 60$ ,  $P = 100 - 60 = 40$
- Profit per firm:  $\pi = (40 - 10) \times 30 = 900$

## Cournot with $n$ symmetric firms: derivation

- Linear demand  $P = a - bQ$  where  $Q = q_1 + q_2 + \dots + q_n$
- Firm  $i$  profit:

$$\pi_i = q_i(a - bQ) - cq_i = q_i(a - bQ - c)$$

- FOC:

$$\frac{d\pi_i}{dq_i} = a - bQ - bq_i - c = 0$$

- By symmetry:  $q_1 = q_2 = \dots = q_n = q^*$ , so  $Q = nq^*$
- Substituting:  $a - bnq^* - bq^* - c = 0$
- Solving:  $q^* = \frac{a-c}{b(n+1)}$ ,  $P^* = \frac{a+nc}{n+1}$
- **Key insight:** As  $n \rightarrow \infty$ ,  $P \rightarrow c$  (perfect competition)

## Practice: Cournot comparative statics

- **True, False, or NEI:**
- (a) Adding a third firm to a Cournot duopoly always reduces industry profits.
- (b) In Cournot, all firms must have the same marginal cost.

## Practice: Cournot comparative statics (solution)

- **Answers:**
- **(a) TRUE.** More firms  $\Rightarrow$  lower price  $\Rightarrow$  lower industry profit. Each firm's profit falls, and total profit falls because  $P$  is closer to  $MC$ .
- **(b) FALSE.** Asymmetric costs work fine. Low-cost firms produce more and earn higher margins.

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2. **Bertrand competition (refresher)**
3. Cournot vs Bertrand

## Bertrand competition: setup

- $n$  firms producing **homogeneous** goods
- Firms choose **prices** simultaneously
- Consumers buy from lowest-price firm
- If tie: split demand equally
- Constant marginal cost:  $c$

## Bertrand: the paradox

- **Nash equilibrium:**  $p_1 = p_2 = c$  (marginal cost pricing!)
- **Why?**
  - If  $p_i > p_j > c$ : firm  $i$  can undercut and capture entire market
  - Undercutting continues until  $p = c$
- **The “paradox”:**
  - Only 2 firms, but competitive outcome!
  - Zero profits with just 2 competitors
  - Seems unrealistic for most markets

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2. Bertrand competition (refresher)
3. **Cournot vs Bertrand**

## Cournot vs Bertrand: summary

|                    | <b>Cournot</b> | <b>Bertrand</b> |
|--------------------|----------------|-----------------|
| Strategic variable | Quantities     | Prices          |
| Equilibrium price  | $P > MC$       | $P = MC$        |
| Profits            | Positive       | Zero            |

- Which model is “right”?
- Answer: depends on the industry!

# Escaping the Bertrand paradox

## 1. **Capacity constraints** (Kreps-Scheinkman)

- Can't serve entire market at low price
- Leads to Cournot outcome

## 2. **Product differentiation** (next lecture!)

- Consumers have preferences
- Not all switch to lowest price

## 3. **Repeated interaction** (collusion, covered later)

- Firms can sustain  $P > MC$  through punishment strategies

## When does each model apply?

- **Cournot applies when:**
  - Capacity constraints matter
  - Firms commit to production before selling
  - Quantities are hard to adjust quickly
- Examples: manufacturing, airlines (seat capacity)
- **Bertrand applies when:**
  - Prices adjust quickly
  - No capacity constraints
  - Homogeneous products
- Examples: online retail, commodities

## Industry examples: which model?

| Industry          | Model          | Why?                                  |
|-------------------|----------------|---------------------------------------|
| Airlines          | Cournot        | Capacity (planes, gates) committed    |
| Cement            | Cournot        | Production committed; transport costs |
| Gasoline stations | Bertrand       | Prices adjust daily; commodity        |
| Online retail     | Bertrand       | Instant price changes; no capacity    |
| Smartphones       | Differentiated | Product differentiation dominates     |

- In practice: most markets have differentiated products!

## Practice: Cournot vs Bertrand

- **Question:** Two firms have  $MC = 20$ . Market demand is  $P = 100 - Q$ .
- (a) Find equilibrium price under Cournot.
- (b) Find equilibrium price under Bertrand.
- (c) Which model yields higher consumer surplus? Why?

## Practice: Cournot vs Bertrand (solution)

- **Solution:**

- **(a) Cournot:** Using  $P^* = \frac{a+nc}{n+1} = \frac{100+2(20)}{3} = 46.67$

- Or:  $q^* = \frac{100-20}{3} = 26.67$ ,  $Q = 53.33$ ,  $P = 46.67$

- **(b) Bertrand:**  $P = MC = 20$

- **(c)** Bertrand has higher CS:

- Bertrand:  $CS = \frac{1}{2}(100 - 20)(80) = 3200$

- Cournot:  $CS = \frac{1}{2}(100 - 46.67)(53.33) \approx 1422$

- Lower price  $\Rightarrow$  higher quantity  $\Rightarrow$  higher CS

## Kreps-Scheinkman (1983): resolving the puzzle

- Two-stage game:
  1. Stage 1: Firms choose capacities (quantities)
  2. Stage 2: Firms compete in prices
- **Result:** Equilibrium outcome = Cournot!
- **Intuition:**
  - Capacity choice commits firms
  - Price competition is constrained by capacity
  - Undercutting is limited by what you can produce
- Key insight: commitment matters

## Oligopoly refresher: Key Points

1. **Cournot:** Firms choose quantities;  $P > MC$
2. **Bertrand (homogeneous):**  $P = MC$ , zero profits
3. **Bertrand paradox:** Only 2 firms but competitive outcome
4. Cournot applies with capacity constraints; Bertrand with flexible prices
5. **Escaping the paradox:** Capacity constraints, differentiation, collusion
6. **Kreps-Scheinkman:** Capacity then price  $\Rightarrow$  Cournot outcome

## Next time

- **Lecture 8:** Product Differentiation and Hotelling
  - Product differentiation: why it matters
  - Hotelling model of spatial competition
  - Connection to demand estimation